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MILITARY SPECIFICATION

CONVERTERS, LIQUID OXYGEN, CAPACITANCE TYPE GAGING, GENERAL SPECIFICATION FOR

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for liquid oxygen converters that will store liquid oxygen on board an aircraft and that will convert the liquid oxygen to gas as required to supply the oxygen regulators. It includes converters of any specified operating pressure with either integral warming coils or separate aircraft installed warming coils and with either provisions for quick-change installation in aircraft or provisions for permanent installation in aircraft.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

BB-N-411	Nitrogen, Technical
QQ-S-766	Steel Plates, Sheets, and Strip, Corrosion Resisting

MILITARY

MIL-P-116	Preservation-Packaging, Methods of
MIL-S-4043	Steel, Corrosion Resisting (Extra Low Carbon Type 304) Plate, Sheet and Strip
MIL-P-7105	Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT, General Requirements for
MIL-S-7742	Screw Threads, Standard Optimum Selected Series: General Requirements for
MIL-V-9050	Valves, Oxygen, Pressure Relief, Aircraft

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: ASD/ENESS, Wright-Patterson AFB, OH 45433 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-F-18264	Finishes: Organic, Weapons System, Application and Control of
MIL-D-19326	Design and Installation of Liquid Oxygen Systems in Aircraft, General Specification for
MIL-C-25516	Connectors, Electrical, Miniature, Coaxial, Environment Resistant Type, General Specification for
MIL-L-25567	Leak Detection Compound, Oxygen Systems
MIL-I-25645	Indicator, Liquid Oxygen Quantity, Capacitance Type, General Specification for
MIL-V-25961	Valve, Fill-Buildup-Vent, Liquid Oxygen Converter CRU-50/A
MIL-O-27210	Oxygen, Aviator's Breathing, Liquid and Gas
MIL-T-27602	Trichloroethylene, Oxygen Propellant Compatible
MIL-T-27730	Tape, Antiseize, Polytetrafluoroethylene, With Dispenser
MIL-V-38201	Valve, Filler, Liquid Oxygen, Female CRU-59/E
MIL-C-81302	Cleaning Compound, Solvent, Trichlorotrifluoroethane
MIL-T-81533	1, 1, 1 Trichloroethane (Methyl Chloroform) Inhibited, Vapor Degreasing

STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

DOD-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-721	Definitions of Effectiveness Terms for Reliability, Maintainability, Human Factors, and Safety
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-889	Dissimilar Metals
MS33583	Tubing End, Double Flare, Standard Dimensions for
MS33584	Tubing End, Standard Dimensions for Flared

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specificaton to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION (NASA)

NHB 8060.1	Flammability, Odor, and Offgassing Requirements and Test Procedures for Materials in Environments that Support Combustion

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(Copies of this publication can be obtained from the National Aeronautics and Space Administration Headquarters; Office of Reliability and Quality Assurance (Code KR); Washington, D.C. 20546.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 2512 Compatibility of Materials with Liquid Oxygen (Impact Sensitivity Threshold Technique)

(Copies of this publication can be obtained from the American Society for Testing and Materials; 1916 Race Street; Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available from libraries and are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Major components. The major components of the converters shall be as follows:

(a) Liquid oxygen container

(b) Content sensing element and connections

(c) Fill-buildup-vent valve (applicable to converters with provisions for quick-change installations)

(d) Pressure closing valve

(e) Pressure relief valve

(f) Overpressure safety device

(g) Mounting provisions and system connections.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials. Materials shall conform to applicable specifications as identified herein. Materials that are not covered by specifications shall be the best quality, shall be the lightest practical weight, and shall be suitable for the intended purpose.

3.3.1 Metals. Metals shall be corrosion-resistant steel or nonferrous metal that has been treated to resist corrosion.

3.3.1.1 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals as defined in MIL-STD-889 shall not be used in intimate contact.

3.3.1.2 Protective treatment. Metallic materials that are subject to corrosion in oxygen, salt air, or any other atmospheric conditions likely to

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occur during the usage of the converter shall be protected against such corrosion in a manner that will not in any way prevent compliance with the performance requirements of this specification.

* 3.3.1.3 Nonmetallic materials. Nonmetallic materials shall have the physical properties required for the converter to function, without being damaged and without being adversely affected, while exposed to temperatures from plus 260 degrees Fahrenheit to minus 297 degrees Fahrenheit. The contractor shall be responsible for the evaluation of materials for their suitability in each application. The contractor may evaluate materials in accordance with either ASTM D 2512 or NASA NHB 8060.1.

3.4 Design and construction. Each converter shall be designed and constructed as specified herein and in the applicable detail specification.

3.4.1 Tubing and fittings. All openings of the finished converter shall be capped or plugged to prevent thread damage and to prevent the entry of moisture and foreign matter. Aluminum tubing that interconnects components shall be double flared to conform to MS33583. Tube fittings and pipe thread bosses for mating the converter to aircraft lines shall be corrosion-resistant steel. Corrosion-resistant steel tubing shall be single flared to conform to MS33584.

3.4.2 Antiseize tape. Antiseize tape that conforms to MIL-T-27730 shall be used on tapered male pipe thread fittings. The antiseize tape shall not be used on the straight threads or the tubing fittings. Particles of the antiseize tape shall not be allowed to enter the inside of plumbing lines.

3.4.3 Check valves and flow restrictors. Check valves and flow restrictors may be used in the circuit as required for function. Check valves and flow restrictors shall be spring loaded as necessary to avoid any possibility of entrapment of liquid oxygen in a dead end circuit.

3.4.4 Threads. Screw threads shall be in accordance with MIL-S-7742. Pipe threads shall be in accordance with MIL-P-7105.

3.4.5 Cleaning

* 3.4.5.1 Preassembly of converter. Prior to the assembly of the converter, the internal and external surfaces of the components shall be degreased by ultrasonics, by immersing and scrubbing, by pressure spraying with a cleaning compound that conforms to MIL-C-81302, or by vapor degreasing with a solvent that conforms to MIL-T-27602, type I or II of MIL-C-81302, or MIL-T-81533. After the components have been cleaned, a General Electric Co. Type H Leak Detector, or equivalent halide testing apparatus, shall be used to determine that the cleaning compounds have not been retained.

* 3.4.5.2 After welding and brazing. After all welding and brazing operations have been completed, all weld areas shall be cleaned, and weld spatter shall be removed by suitable mechanical or abrasive cleaning techniques. All flux residue shall be thoroughly removed in order to preclude corrosive effects which may occur undetected beneath the paint and result in early failure of the converter in service. The cleaning solutions and the processes that are used shall be compatible with the converter and its expected service life.

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3.5 Performance

* 3.5.1 Final assembly pressure. When the converter is subjected to the final assembly pressure specified in the applicable detail specification, it shall not leak except for the leakages allowed for installed components by their respective specifications. The specified primary pressure relief valve shall vent the converter at the flow rate established in MIL-V-9050 in response to an internal pressure not greater than 10 pounds per square inch gage above the maximum normal working pressure of that valve. The primary relief valve is the component relief valve that has the lowest operating pressure range.

3.5.2 Filling time. When liquid oxygen is transferred from standard equipment to the converter, the time required to fill the converter shall not exceed the filling time specified in the applicable detail specification.

3.5.3 Evaporation loss

3.5.3.1 Vent condition. The evaporation loss of the essentially full converter in the vent condition shall not exceed the maximum evaporation loss specified in the applicable detail specification.

3.5.3.2 Buildup condition. The evaporation loss of the essentially filled converter in the buildup condition shall not exceed the maximum evaporation loss specified in the applicable detail specification.

3.5.4 Buildup time. The time required for the converter to reach its operating pressure immediately after it has been filled shall not exceed the buildup time specified in the applicable detail specification.

3.5.5 Dielectric strength. When the dielectric strength of the sensing element in the empty converter is measured at a potential of 500 volts rms, 60 Hertz, the current leakage shall not exceed 1 milliampere.

3.5.6 Insulation resistance. When a potential of 50 volts direct current is applied between the two lead wires of the sensing element in the empty converter, the insulation resistance shall not be less than 20 megohms. When a potential of 50 volts direct current is applied between each lead wire of the sensing element in the empty converter and the converter ground, the insulation resistance between either lead wire and the converter ground shall not be less than 10 megohms.

3.5.7 Capacitance. The capacitance of the sensing element, after installation in the empty converter, shall be within the limits specified in the applicable detail specification.

* 3.5.8 Permanent volumetric expansion. The permanent volumetric expansion of the inner vessel shall not exceed 10 percent of the total volumetric expansion while the inner vessel is subjected to the pressure specified in the applicable detail specification.

3.5.9 Pressure closing valve leakage. When the converter buildup pressure conditions are applied to the pressure closing valve, the leakage past the valve seat shall not exceed 0.02 liter per minute measured at normal temperature and pressure, with a pressure diffential of 5 to 10 pounds per square inch being maintained across the pressure closing valve seat.

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3.5.10 Delivery rate. The converter shall deliver oxygen at both a constant flow rate and cyclically to provide the mass flow and delivery pressure specified in the applicable detail specification.

3.5.11 Calibration. When liquid oxygen in approximately equal increments, measured by weight, is removed from the filled converter, the capacitance values shall establish an essentially constant slope and shall be within the range specified in the applicable detail specification.

3.5.12 Operating temperature. The converter shall function as specified in 3.5.10 at any ambient temperature in the range of minus 65 degrees Fahrenheit to plus 260 degrees Fahrenheit.

3.5.13 Overpressure safety. The overpressure safety device shall remain leaktight under all normal use conditions. The overpressure safety device shall open at any pressure level between 1.4 and 2.5 times the maximum opening pressure of the primary pressure relief valve but in all cases at a pressure low enough to preclude failure of the liquid oxygen container.

3.5.14 Rupture area. The rupture area in the outer vessel of the liquid oxygen container shall open in response to an internal pressure that is not more than 50 percent of the burst pressure of the outer vessel. The rupture area shall be capable of forming an opening that is equivalent to a hole with a diameter of at least 3/8 inch.

3.5.15 Vibration. The converter, during and after exposure to vibration, shall function within the limits established in the applicable detail specification.

3.5.16 Acceleration. After the filled converter has been subjected to acceleration, the converter shall meet the requirements specified in 3.5.1 and 3.5.3.1.

3.5.17 Shock. After the filled converter has been subjected to shock, the converter shall meet the requirements specified in 3.5.1 and 3.5.3.1.

3.5.18 Crash safety. When the filled converter is mounted to simulate a typical aircraft installation and subjected to crash loads, the converter shall not break free from the mounting provisions.

3.5.19 Salt fog. Exposure of the converter to salt fog shall not cause corrosion indicative of dissimilar metals or corrosion that could preclude normal operation of the converter or create any unsafe conditions. After the salt deposits are removed from the converter, the converter shall meet the requirements specified in 3.5.1, 3.5.3.2, 3.5.5, 3.5.6, 3.5.7, and 3.5.11.

3.5.20 Gunfire. When the section of the converter that contains the liquid oxygen is pierced by a gun-fired projectile, the converter shall not shatter and shall remain in one piece except for small pieces torn from or broken off the converter by the projectile. The mounting provisions of the converter shall withstand the forces of impact of the gun-fired projectile and the discharge of the contents of the converter.

3.5.21 Reliability. When the converters are operated in a manner simulating actual use in aircraft, the converters shall demonstrate a mean time between failures (MTBF) of not less than 300 hours.

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3.5.21.1 Service life. The service life of the converters, which is the period of use during which the MTBF remains essentially constant, shall be not less than 1,000 operational hours, excluding standby period when no flow is being taken from the supply connection.

3.5.22 Maintainability. The corrective maintenance as defined in MIL-STD-721 shall not exceed 2 hours mean maintenance time to repair the plumbing leaks; to remove and replace the components such as the pressure closing valve, the relief valve, or the overpressure safety device; or to replace the tubing runs and the mounting hardware. The loss of vacuum or capacitance gage malfunctions, which require the replacement of the entire liquid oxygen container, shall be excluded from this requirement.

3.6 Details of components

3.6.1 Liquid oxygen container. The liquid oxygen container shall be a double-walled, vacuum-insulated container that will limit heat transfer to the minimum and maintain a near perfect permanent vacuum. The vacuum between the inner wall (inner vessel) and the outer wall (outer vessel) shall contain a getter that will not propagate fire or explosion in the event of a direct hit by an incendiary projectile. The fill line entrance to the storage space of a liquid oxygen container shall not allow the filling stream to impinge on the vent outlet opening. The inner vessel and the outer vessel shall be fabricated utilizing manufacturing processes and selected corrosion-resistant steels that conform to QQ-S-766 or MIL-S-4043 to assure limited chromium carbide grain formulation thereby reducing susceptibility to stress corrosion. The outer vessel shall have a rupture area that is positioned or shielded to minimize the danger to nearby personnel.

* 3.6.2 Content sensing element and connections. A content sensing element that will provide a linear incremental increase in electrical capacitance proportional to the liquid oxygen quantity shall be incorporated within the liquid oxygen container. The content sensing element shall function with gage system components that conforms to MIL-I-25645. The two leads from the content sensing element shall terminate in electrical receptacles. The electrical receptacle for high impedance lead shall conform to style B of MIL-C-25516. The electrical receptacle for the low impedance lead shall conform to style E of MIL-C-25516. The electrical receptacles of the converters shall be provided with storage caps that will preclude contamination of the receptacle terminals during transit and storage.

3.6.3 Fill-buildup-vent valve. When the applicable detail specification specifies a fill-buildup-vent valve as a component of the converter, the fill-buildup-vent valve shall meet the performance requirements specified in MIL-V-25961. The fill-buildup-vent valve shall be located so that it will interface and provide the hand operation clearance necessary to accomplish normal filling actions with a CRU-59/E female filler valve that conforms to MIL-V-38201.

* 3.6.4 Pressure closing valve. The pressure closing valve shall be actuated by a leakproof metal bellows as the pressure responsive element. The adjustment feature of the pressure closing valve shall be secured in place by safety wire or sealant, or both, to prevent vibration induced changes.

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* 3.6.5 Pressure relief valve. The pressure relief valves shall meet the performance requirements of MIL-V-9050. The type of pressure relief valve or valves identified in the applicable detail specification shall be utilized. The maximum normal working pressure for the converter shall be the maximum of the pressure range for the primary pressure relief valve.

3.6.6 Overpressure safety device. A frangible safety disc or other secondary overpressure device shall be incorporated in the plumbing of the converter at a location to insure rapid dumping of the head pressure from the converter in the event the pressure relief valve fails to open.

3.6.7 Mounting provisions and system connections. The mounting provisions for either quick-change or fixed installation in aircraft and system connections for the converter shall be as specified in the applicable detail specification.

3.7 Interchangeability. All parts having the same manufacturer's part numbers shall be dimensionally and functionally interchangeable. The item identification and part number requirements of DOD-STD-100 shall apply.

3.8 Dimensions. The dimensions of the converter shall be as specified in the applicable detail specification.

3.9 Weight. The weight of the complete converter, including the converter half of any quick disconnects, shall not exceed the weight specified in the applicable detail specification.

* 3.10 Finish. After the cleaning specified in 3.4.5.1 and 3.4.5.2 has been completed, the assembled liquid oxygen container and any parts of the other components that are to be painted shall be coated with a primer compatible with the finish to be applied. One or more coats of durable, high quality paint shall be applied as required to achieve a smooth protective finish free of cracks, checks, or runs. The color of the topcoat shall conform to 14187 of FED-STD-595. Organic finishes shall be applied in accordance with MIL-F-18264.

3.11 Marking

* 3.11.1 Identification of product. Each converter shall be permanently identified as specified in MIL-STD-130.

* 3.11.2 Precautionary marking. Each converter shall have a decal or a plate that is located in a conspicuous place. The following information shall be marked on the decal or plate.

WARNING
HIGH VACUUM CONTAINER
HANDLE WITH CARE

3.12 Workmanship. The converters shall be manufactured in accordance with high grade aircraft equipment manufacturing practices. The converters shall be thoroughly cleaned of oil and any other foreign matter and shall be free of defects that might affect appearance or functionability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection conditions

4.2.1 Environmental conditions. Unless otherwise specified, all tests shall be conducted at local ambient temperature and barometric pressure. The temperature and barometric pressure at the time each test is accomplished shall be recorded. This information shall be available for computation of corrections, where required, to normal temperature and pressure conditions. Normal temperature and pressure conditions are 14.7 pounds per square inch absolute and plus 70 degrees Fahrenheit. Test instruments shall be calibrated or adjusted according to their required usage in conducting the tests specified herein.

4.2.2 Master capacitance bridge. The master capacitance bridge that is used to test the sensing element shall have an accuracy equal to or better than plus or minus 0.2 percent of the nominal full converter capacitance or 0.2 picofarad, whichever is larger. The accuracy of the master capacitance bridge used for the measurements specified herein shall be applied to broaden the acceptable capacitance range specified in the applicable detail specification.

4.2.3 Cryogenic liquids and gases. The gaseous or liquid oxygen used for the tests specified herein shall conform to MIL-O-27210. If the use of oxygen for the tests specified herein is hazardous because of the nature of the test area or equipment, liquid or gaseous nitrogen that conforms to BB-N-411 may be used.

4.2.4 Post test cleanup. Upon the conclusion of testing in an unclean area any foreign material contamination shall be removed before resuming testing with oxygen. All traces of leak detection compound shall be removed from the converter after each leakage test.

4.2.5 Purging. After the inspection of the converters has been completed, each converter shall be purged with hot dry gas. The gas used for purging shall be either nitrogen that conforms to BB-N-411 or oxygen that conforms to MIL-O-27210. The temperature of the purging gas at the inlet of the converter shall not exceed plus 250 degrees Fahrenheit.

4.3 Classification of inspection. The classification of inspection (see 6.3) shall be as specified in the applicable detail specification and shall include the quality conformance inspection described herein.

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4.4 Quality conformance inspection. The quality conformance inspection shall consist of the following:

(a) Individual inspection

(b) Sampling inspection.

4.4.1 Individual inspection

4.4.1.1 Converters. Each converter shall be subjected to the following methods of inspection:

(a) Visual examination (see 4.6.1.1)

(b) Final assembly pressure (see 4.6.2)

(c) Filling time (see 4.6.3)

(d) Evaporation loss (see 4.6.4 through 4.6.4.2)

(e) Buildup time (see 4.6.5)

(f) Dielectric strength (see 4.6.6)

(g) Insulation resistance (see 4.6.7)

(h) Capacitance (see 4.6.8).

* 4.4.1.2 Inner vessels. Before the outer vessel is added to the inner vessel, each assembled inner vessel complete with all lines and fittings shall be subjected to the permanent volumetric expansion test described in 4.6.9.

4.4.1.3 Pressure closing valves. Before the pressure closing valves are installed in the buildup circuit of the converters, each pressure closing valve shall be subjected to the pressure closing valve leakage test described in 4.6.10.

4.4.2 Sampling inspection. The lot size for plans A, B, C, and D shall be 100 sample units that have been manufactured under essentially the same conditions and submitted for inspection at essentially the same time. The unit for the expression of the lot size and the sample unit for plans A and B shall be one complete converter. The unit for the expression of the lot size and the sample unit for plan C shall be one overpressure safety device before it has been installed in the buildup circuit of the converter. The unit for the expression of the lot size and the sample unit for plan D shall be components for one outer shell.

4.4.2.1 Plan A. Five converters shall be selected at random from each lot or fraction thereof and shall be subjected to the following methods of inspection:

(a) Dimensional examination (see 4.6.1.2)

(b) Delivery rate (see 4.6.11.1 and 4.6.11.2)

(c) Calibration (see 4.6.12).

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4.4.2.2 Plan B. One converter shall be selected at random from each lot or fraction thereof and shall be subjected to the operating temperature tests described in 4.6.13.1 and 4.6.13.2.

4.4.2.3 Plan C. One overpressure safety device shall be selected at random from each lot or fraction thereof and shall be subjected to the overpressure safety test described in 4.6.14.

4.4.2.4 Plan D. The components for one outer vessel shall be selected at random from each lot or fraction thereof and shall be assembled. The assembled outer shell shall be subjected to the rupture area test described in 4.6.15.

4.4.2.5 Rejection and retest. If one or more items selected from a lot fail to meet the specification, acceptance of all items in the lot shall be withheld until the extent and cause of failure have been determined. The contractor shall explain fully to the Government representative and notify the procuring activity in writing the cause of failure, the action taken to preclude recurrence, and the impact this failure might have on scheduled deliveries. After correction, all of the sampling inspection shall be repeated.

4.4.2.6 Continuation of inspection. For production reasons, individual inspection or other sampling plans may be continued pending the investigation of a sampling inspection failure. Final acceptance of the entire lot or lots produced later shall not be made until it is determined that all items meet all the requirements of this specification.

4.4.3 Defects in items already accepted. The investigation of a test failure could indicate that defects might exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and the method of correcting them.

4.5 Inspection of preparation for delivery. Shipping containers fully prepared for delivery except for closure shall be examined and tested, in accordance with MIL-STD-794, to determine conformance to the requirements for preservation, packaging, packing, and marking.

4.6 Methods of inspection

4.6.1 Examination of product

4.6.1.1 Visual examination. The converter shall be visually examined to determine conformance to the requirements of this specification and the applicable detail specification with respect to materials, configuration, threads, weight, finish, marking, and workmanship. Unless otherwise specified herein, nonconformance of the converter to any requirement specified with respect to materials, configuration, threads, weight, finish, marking, and workmanship shall be cause for rejection of the converter. Five defects per 100 converters will be acceptable for appearance defects, which are not detrimental to function, such as abnormal tool marks, poor quality finish, tubing contour, scrapes, or scratches.

4.6.1.2 Dimensional examination. The converter shall be examined to determine conformance to the dimensions specified in the applicable detail specification. If any dimension of the converter is not as specified, the converter shall be rejected.

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4.6.2 Final assembly pressure. The converter, including all its fittings and connections, shall be subjected to the pneumatic pressure that is specified as the final assembly pressure in the applicable detail specification. While this pressure is maintained, the converter shall be tested by applying leak detection compound that conforms to MIL-L-25567 to all fittings and connections of the converter and then examining all fittings and connections of the converter for leakage. The pressure shall then be increased to a level sufficient to open the pressure relief valve and to demonstrate that the required flow capacity can be achieved without excessive pressure. If there is any leakage from the converter, except for the leakages that are allowed for installed components by their respective specifications, the converter shall be rejected.

4.6.3 Filling time. The converter shall be stabilized at approximately plus 70 degrees Fahreheit. Liquid oxygen shall be transferred from a cryogenic vessel, through a precooled liquid transfer hose and a CRU-59/E female filler valve that conforms to MIL-V-38201, to the converter. If a fill-buildup-vent valve is a component of the converter, the inlet pressure of the converter shall be maintained at 30 pounds per square inch. If a fill-buildup-vent valve is not a component of the converter, the inlet pressure of the converter shall be maintained at 20 pounds per square inch. The liquid oxygen shall be transferred until the weight gain of the converter is equivalent to a full charge of liquid oxygen which shall be the minimum capacity that is specified for the converter in the applicable detail specification. If the time required to fill the converter exceeds the filling time specified in the applicable detail specification, the converter shall be rejected.

4.6.4 Evaporation loss

* 4.6.4.1 Vent condition. The evaporation loss in the vent condition for a 24-hour period shall be determined two times as specified herein. The first (initial) determination may be made anytime after evacuation and pinch-off of the liquid oxygen container. The second determination shall be made after a lapse of at least 30 days from the date of the first determination. If the first or the second determination exceeds the evaporation loss specified in the applicable detail specification, the liquid oxygen container shall be rejected. The evaporation rate shall be determined as follows: The liquid oxygen container shall be filled with liquid oxygen and allowed to stabilize for 2 hours in the vent condition and then weighed (initial weight measurement). At least 24 hours after the initial weight measurement has been made, the liquid oxygen container shall be reweighed, and the evaporation loss in the vent condition (difference in weight) shall be determined.

4.6.4.2 Buildup condition. The evaporation loss in the buildup condition for a 24-hour period shall be determined as follows: The converter shall be filled with liquid oxygen and then placed in the buildup condition. The converter shall be weighed (initial weight measurement) as soon as operating pressure has developed. At least 24 hours after the initial weight measurement has been made, the converter shall be weighed again, and the evaporation loss in the buildup condition (difference in weight) shall be determined. If the evaporation loss in the buildup condition exceeds the evaporation loss value specified in the applicable detail specification, the converter shall be rejected.

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4.6.5 Buildup time. The converter shall be filled with liquid oxygen and, without delay, shall be set up to generate the operating pressure specified in the applicable detail specification. If the time required for the converter to reach its normal operating pressure range exceeds the buildup time specified in the applicable detail specification, the converter shall be rejected.

4.6.6 Dielectric strength. The dielectric strength of the sensing element shall be measured at a potential of 500 volts rms, 60 Hertz. This potential shall be maintained for a period of 10 seconds between each lead wire and the converter ground and between the two lead wires. If the current leakage measured at this potential exceeds 1 milliampere, the converter shall be rejected.

4.6.7 Insulation resistance. The insulation resistance of the sensing element shall be measured at a potential of 50 volts direct current. This potential shall be applied between each lead wire and the converter ground and between the two lead wires. If the insulation resistance measured between the two lead wires is less than 20 megohms or if the insulation resistance measured between either lead wire and the converter ground is less than 10 megohms, the converter shall be rejected.

4.6.8 Capacitance. With the converter empty, equilized at room temperature, and dry, the capacitance of the sensing element shall be measured with the master capacitance bridge. If the capacitance of the sensing element is not within the range specified in the applicable detail specification, the converter shall be rejected.

* 4.6.9 Permanent volumetric expansion. Any internal pressure that has been applied to the inner vessel before the permanent volumetric expansion test shall not have exceeded 90 percent of the pressure to be applied during this test. The assembled inner vessel, complete with all lines and fittings, shall be enclosed in a water jacket where the inner vessel will be totally submerged and free to expand in all directions. The inner vessel shall be subjected to the hydrostatic or pneumatic pressure specified in the applicable detail specification for at least 1 minute, or longer, to insure the complete expansion of the inner vessel. The difference in temperature of the fluid inside the inner vessel and the fluid outside the inner vessel shall not exceed 5 degrees Fahrenheit. If the test pressure cannot be maintained due to failure of the test apparatus, the permanent volumetric expansion test may be repeated with the test pressure increased by 10 percent. If the permanent volumetric expansion of the inner vessel exceeds 10 percent of the total volumetric expansion that existed while the inner vessel retained the specified test pressure, the inner vessel shall be rejected.

* 4.6.10 Pressure closing valve leakage. A pneumatic pressure source shall be applied to the inlet of the pressure closing valve. Suitable pressure gages and control valves shall be installed in the inlet and outlet of the pressure closing valve. Pressure shall be applied gradually to the inlet of the pressure closing valve until the pressure closing valve closes. The inlet pressure shall be gradually increased until it exceeds the closing pressure by 5 to 10 pounds per square inch gage. The leakage measured across the valve seat should not exceed 0.02 liter per minute at normal temperature and pressure. If the leakage is greater than 0.02 liter per minute, the pressure closing valve shall be rejected.

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4.6.11 Delivery rate

* 4.6.11.1 Steady flow. The converter shall be set up to simulate installation in an aircraft. If the converter does not have integral warming coils, the tubing specified in the table entitled "Minimum Length and Tubing Between Converters and First Station" of MIL-D-19326, or an equivalent heat exchanger, shall be added to the supply line. The supply line shall be connected to a metering valve and a suitable flow monitoring instrument, which can be used to observe and set the flow specified in the applicable detail specification. The converter shall be filled, set for pressure buildup, and then operated until empty. If the converter does not deliver the maximum steady flow specified in 3.5.10, after allowing 10 minutes for stabilization, the converter shall be rejected. Momentary pressure drops below the specified normal operating pressure during the first 10 minutes of flow will be acceptable.

* 4.6.11.2 Cyclic flow. The converter shall be set up to simulate installation in an aircraft. If the converter does not have integral warming coils, the tubing specified in the table entitled "Minimum Length and Tubing Between Converters and First Station" of MIL-D-19326, or an equivalent heat exchanger, shall be added to the supply line. The supply line shall be connected to a cyclic flow control mechanism and a suitable flow monitoring system, which can be adjusted to obtain the required cyclic flow. The converter shall be filled with liquid oxygen, set for pressure buildup, and then operated until empty. If the converter does not deliver the oxygen at the average flow rate specified in the applicable detail specification, after allowing 10 minutes for stabilization, the converter shall be rejected. Momentary pressure drops below the specified normal operating pressure during the first 10 minutes of flow will be acceptable.

4.6.12 Calibration. The converter shall be filled to capacity with liquid oxygen. After the filled converter has been stabilized for a few minutes, the capacitance of the sensing element shall be measured by the master capacitance bridge. Five essentially equal increments of liquid oxygen shall be removed from the converter. After the removal of each increment of liquid oxygen from the converter, the capacitance of the sensing element shall be measured with the master capacitance bridge. If the capacitance values measured do not remain within the range specified in the applicable detail specification, the converter shall be rejected.

4.6.13 Operating temperature

4.6.13.1 Low temperature. The converter shall be installed in an environmental temperature chamber and connected into a simulated aircraft oxygen system. The converter shall be filled with liquid oxygen or liquid nitrogen and conditioned at minus 65 degrees (plus or minus 2 degrees) Fahrenheit for at least 3 hours. Flow cycling on and off at the rate and volume to total 80 to 110 percent of the maximum flow required by the applicable detail specification shall be initiated and continued until the converter is within 5 percent of empty. If the converter does not deliver the flow of oxygen specified above, after allowing 10 minutes for stabilization, the converter shall be rejected. Momentary pressure drops below the specified normal operating pressure during the first 10 minutes of flow shall be acceptable.

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4.6.13.2 High temperature. The converter shall be installed in an environmental temperature chamber and connected into a simulated aircraft oxygen system. The converter shall be filled with liquid oxygen or liquid nitrogen and conditioned at plus 260 degrees (plus or minus 2 degrees) Fahrenheit for at least 3 hours. Flow cycling on and off at the rate and volume to total 80 to 110 percent of the maximum flow required by the applicable detail specification shall be initiated and continued until the converter is within 5 percent of empty. If the converter does not deliver the flow of oxygen specified above, after allowing 10 minutes for stabilization, the converter shall be rejected. Momentary pressure drops below the specified normal operating pressure during the first 10 minutes of flow will be acceptable.

* 4.6.14 Overpressure safety. The overpressure safety device shall be subjected to gradually increasing pressure. If the overpressure safety device leaks at a pressure below 1.4 times the maximum normal working pressure (the pressure range specified in MIL-V-9050 for the primary pressure relief valve) or fails to open at 2.5 times the maximum normal working pressure, the overpressure safety device shall be rejected.

4.6.15 Rupture area. The assembled outer vessel complete with all lines and fittings shall be pressurized until it fails. If failure does not occur in the rupture area and in a controlled manner to satisfy the requirements specified in 3.5.14, the components of the outer vessel shall be rejected.

* 4.6.16 Vibration. The converter shall be evaluated in accordance with the guidelines described under "Performance of Test" in section 3 of MIL-STD-810. The converter shall be tested in accordance with method 514, procedure IA, figure entitled "Random Vibration Test Curve of Mass Loading Reduction Factor for Jet Aircraft Equipment" of MIL-STD-810 unless otherwise specified herein. W_o shall be 0.30 G^2/Hertz. The test time shall be 1 hour along each perpendicular axis. In lieu of actual operation with liquid oxygen or liquid nitrogen, the converter may be filled with a fluid to simulate the operating mass and pressurized to the normal operating conditions throughout the vibration exposure. At least 48 hours after the converter has been subjected to the vibration test, the converter shall be subjected to and shall pass the tests described in 4.6.2 and 4.6.4.2.

* 4.6.17 Acceleration. The converter shall be filled, allowed to reach operating pressure under no flow conditions, and then subjected to an acceleration test in accordance with method 513, procedure II of MIL-STD-810. In lieu of actual operation with liquid oxygen or liquid nitrogen, the converter may be filled with a fluid to simulate the operating mass and pressurized to the normal operating condition during the acceleration exposure. The forward-backward and lateral forces applied shall be 4 G since mounting orientation is not standardized. The vertically downward force applied shall be 9 G. The vertically upward force applied shall be 3 G. After the converter has been subject to the acceleration test, the converter shall be subjected to and shall pass the tests described in 4.6.2 and 4.6.4.2

* 4.6.18 Shock. The converter shall be filled, allowed to reach operating pressure under no flow conditions, and then subjected to a shock test in accordance with method 516, procedure I, table entitled "Half Sine Shock Pulse Configuration and Its Tolerance Limits" of MIL-STD-810 except that the shock loads shall be determined from "Peak Value (A)" equals 15 G. In lieu of actual

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operation with liquid oxygen or liquid nitrogen, the converter may be filled with a fluid to simulate the operating mass and pressurized to the normal operating condition during the shock exposure. After the converter has been subjected to the shock test, the converter shall be subject to and shall pass the tests described in 4.6.2 and 4.6.4.2.

* 4.6.19 Crash safety. The converter shall be filled with liquid oxygen or liquid nitrogen or loaded to simulate a filled condition, mounted to simulate a typical aircraft installation, and then shall be subjected to a crash safety test in accordance with method 516, procedure III, table entitled "Half Sine Shock Pulse Configuration and Its Tolerance Limits" of MIL-STD-810. If the converter breaks free from its mounting provisions during this test, the converter shall be rejected.

4.6.20 Salt fog. All ports and openings in the liquid oxygen container shall be sealed. The contents receptacles shall be capped. The converter shall then be subjected to a salt fog test in accordance with method 509, procedure I of MIL-STD-810 and then examined for corrosion. If the converter exhibits any corrosion indicative of dissimilar metals or corrosion that could preclude normal operation or create any unsafe condition, the converter shall be rejected. After the removal of any surface salt deposits from the converter, the converter shall be subjected to and pass the tests described in 4.6.2, 4.6.4.2, 4.6.6, 4.6.7, 4.6.8 and 4.6.12.

4.6.21 Gunfire. The converter shall be mounted upon a support that will withstand the forces induced by the gun-fired projectile. The gun-fired projectile shall be a .50 caliber, armor-piercing, incendiary bullet that has been tumbled. The converter shall be filled with liquid oxygen and allowed to reach buildup pressure. After the buildup pressure has been attained, the bullet shall be fired at the converter from a distance of approximately 50 yards and shall strike the converter in the section that contains liquid oxygen. Any evidence of shattering of the converter, failure of the mounting provisions to withstand the forces of impact and the discharge of the contents of the converter, or failure of the converter to remain in one piece shall be cause for the rejection of the converter. Small pieces torn from or broken off the converter by the bullet will be acceptable.

4.6.22 Reliability. The converters shall be operated as specified in 4.6.11.2 except for the following: The cyclic flow shall be approximately 20 percent of the rated capacity of the converter. The cyclic flow shall be interrupted every 100 hours, and the evaporation loss (buildup condition) test described in 4.6.4.2 and the steady flow test described in 4.6.11.1 shall be performed. A failure shall be identified as any malfunction that causes the converter to be unable to deliver the rated flow, unable to maintain the specified operating pressure, or unable to demonstrate acceptable loss rates while on standby. Purging to remove contaminants shall not be considered to be "failure-repair" action. The accept-reject criteria for the reliability testing shall be in accordance with test plan I (10 percent decision risk) of MIL-STD-781.

4.6.23 Maintaintability. The manufacturer shall demonstrate that it is feasible to accomplish disassembly and replacement of exterior components in depot level maintenance shops. Specific components mounted by threaded fasteners shall be replaceable with properly fitting parts by a trained technician within the period specified in 3.5.22.

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5. PACKAGING

5.1 Preservation and packaging. Unless otherwise specified in the applicable detail specification, each converter shall be pressurized with oxygen to a maximum pressure of 25 pounds per square inch gage. The pressurized converter shall be preserved and packaged in accordance with level A or C as specified (see 6.2).

5.1.1 Level A. Each converter shall be preserved and packaged in accordance with method III of MIL-P-116.

5.1.2 Level C. The converters shall be packaged in a manner that will prevent physical damage or deterioration during transit from the shipping point to the original consignee.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Converters that have been packaged as specified shall be packed in accordance with level A of MIL-STD-794.

5.2.2 Level B. Converters that have been packaged as specified shall be packed in accordance with level B of MIL-STD-794.

5.2.3 Level C. Converters that have been packaged as specified shall be packed in accordance with level C of MIL-STD-794.

5.3 Marking. In addition to any special marking required by the contract, interior and exterior containers shall be marked in accordance with MIL-STD-129. In addition each unit package and each shipping container shall be marked as follows:

CAUTION

DO NOT ALLOW PETROLEUM CONTAMINANTS
OF ANY KIND TO BE USED OR STORED ON
OR ABOUT THESE CONTAINERS.

6. NOTES

6.1 Intended use. The converters covered by this specification are intended for use in storing and pressurizing liquid oxygen in aircraft and subsequently converting the liquid oxygen and delivering it as gaseous oxygen at the rate required during flight.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification and of the applicable detail specification.

(b) Applicable levels of preservation and packaging and packing (see 5.1 and 5.2).

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6.3 Classification of inspection. The tests described in 4.6.16 through 4.6.23 will demonstrate the specified physical durability and reliability of the converters under anticipated environmental exposures and loads. These tests are not specified as quality conformance inspection (see 4.3 and 4.4) but should be specified as either first article inspection or qualification inspection in the applicable detail specification. The converters that have been subjected to these tests shall be identified by prominent marking indicating the items have been subjected to these tests and shall not be included in the quantities of serviceable end items.

6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This marking was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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